

- 1 The continuous random variable X has probability density function f given by

$$f(x) = \begin{cases} \frac{1}{6}(x^{-\frac{1}{3}} - x^{-\frac{2}{3}}) & 1 \leq x \leq 27, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the cumulative distribution function of X . [3]

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The random variable Y is defined by $Y = X^{\frac{1}{3}}$.

- (b) Find the probability density function of Y . [3]

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- (c) Find the exact value of the median of Y . [2]

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